



Koneru Lakshmaiah Education Foundation

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DEPARTMENT OF CSE

PROJECT ABSTRACT SUBMISSION

COURSE:SOA PROGRAMMING AND MICROSERVICES– (24SDCS03)

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TITLE OF THE PROJECT	CivicGuardian AI: An Agentic and Service-Oriented Framework for Autonomous Citizen Complaint Management	
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Abstract:

Rapid urbanization and expanding population density have led to an unprecedented surge in civic grievances, spanning critical urban domains such as solid waste management, road infrastructure decay, potable water disruptions, and electrical hazards. Conventional municipal grievance redressal systems are hindered by manual ticketing pipelines, fragmented legacy portals, and human-in-the-loop dispatching mechanisms. Consequently, these systems suffer from prolonged resolution latency, high administrative overhead, subjective classification errors, and opaque grievance lifecycles, ultimately eroding citizen trust and overwhelming civic administrative bodies.

To resolve these operational bottlenecks, this project presents **CivicGuardian AI**, an intelligent, end-to-end, and service-oriented framework engineered for autonomous citizen complaint ingestion, verification, prioritization, and automated departmental dispatch. The system fundamentally shifts civic administration from passive record-keeping to an active, autonomous agentic workflow driven by Large Language Models (LLMs)—including OpenAI GPT and Google Gemini—orchestrated via the LangChain framework.

The proposed architectural framework is structured across four interconnected phases:

1. **Autonomous Ingestion & Multi-Modal Pre-Processing:** Unstructured citizen complaints—comprising natural language text descriptions, geographic coordinates, and timestamped media—are captured via a responsive, accessible web portal developed with React.js and Tailwind CSS. The client layer implements client-side validation and token sanitization before streaming structured payloads to backend services via RESTful APIs secured with JSON Web Tokens (JWT).
2. **Agentic NLP Reasoning, Intent Extraction & Deduplication:** At the reasoning core, LangChain agents interact with state-of-the-art LLMs (GPT/Gemini) to perform zero-shot intent parsing, automated grievance classification across municipal hierarchies, and dynamic severity scoring. Simultaneously, an embedding-based deduplication engine utilizes vector similarity search against historical records stored in MongoDB to identify, cluster, and link redundant or co-located citizen reports, preventing duplicate dispatching.
3. **Service-Oriented Micro-Routing & Persistence:** A high-throughput service layer implemented via Spring Boot and FastAPI decouples incoming traffic from administrative processing. Verified grievances are autonomously routed to targeted municipal endpoints based on real-time organizational mapping. System state, transactional audit logs, and complaint lifecycles are persisted in a MongoDB NoSQL database to support rapid querying and high write availability.
4. **Real-Time SLA Enforcement & Autonomous Escalation:** The platform incorporates an automated monitoring daemon that tracks resolution milestones against dynamic Service Level Agreements (SLAs). In the event of administrative bottlenecks or overdue tickets, the agentic engine initiates hierarchical escalation protocols, real-time citizen status notifications, and performance tracking analytics.

Version control, continuous integration, and automated deployment pipelines are orchestrated using GitHub, with front-end hosting on Vercel and backend microservices deployed on Render. By transforming municipal triage into an autonomous, data-driven agentic ecosystem, CivicGuardian AI minimizes grievance turnaround times, eliminates manual dispatch errors, and establishes an auditable, transparent paradigm for next-generation smart governance.

Project Architecture/Project Overview:

